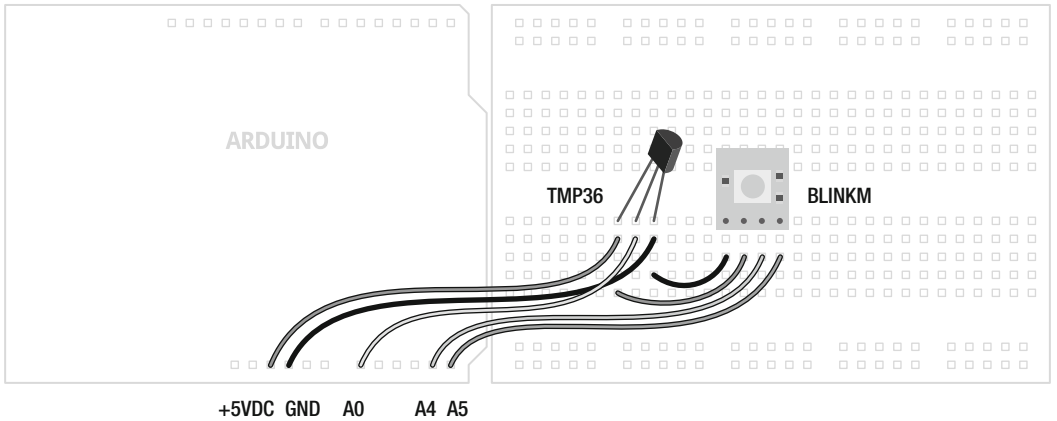


Figure 7-4. *Ambient Temps illustration*

Uploading the Source Code

This sketch for this project will serve as a good example of how to compartmentalize our code by writing and using custom functions, which a good part of this chapter is dedicated to. There are three main parts to this code: reading and calculating the current temperature; mapping that reading to a usable HSB color value; and then sending the appropriate commands to the BlinkM to get it to respond with the color that we want. Let's get this project wired up and the source code in Listing 7-3 uploaded, and continue our discussion from there.

Listing 7-3. *Ambient Temps Source Code*

```
#include <Wire.h>

const int blinkM = 0x09;
const int temperaturePin = A0;

const boolean degreesF = true;

const int hot = 100;
const int cold = 40;
const int hotColor = 0;
const int coldColor = 170;
const int brightness = 255;

void setup() {
  Wire.begin();
  stopScript(blinkM);
  setFadeSpeed(blinkM, 1);
}
```